

Hex3 - An OpenSource Hexapod platform

Manufactured Motion

VIEW IN BROWSER

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Summary

Hex3 is the third generation of our OpenSource Hexapod platform! Control, carry, and expand upon the platform!

Hobby & Makers > RC & Robotics

Tags: robot robotics hexapod

Full documentation is available at https://manufacturedmotion.github.io/Hex/Hex3/#overview
Full project files, including history, is available at https://github.com/manufacturedmotion/hex

Hex3 is the third generation of our Hexapod platform. Highlights of this generation:

- 10kg (~22lb) payload capacity (note: momentary payload capacity may be higher, such as in the images. But your mileage may vary.).
- Drivable via Xbox controller
- Fully 3D printed body and legs
- 18 degrees-of-freedom (DOF) design with 3 DOF per leg
- Custom low-cost servos made using DC gear motors and AS5600 magnetic encoders (as opposed to the hobby-servos that were prone to overheating on Hex2.)

- Spring loaded toe with position sensor to measure force applied by each leg
- Three-level control to follow user commands:
    - High level whole hexapod inverse kinematics
    - Mid level inverse kinematics for each leg
    - Low level control based on cascaded position-velocity controller with feedforward for each axis
- Custom PCBs for Pi and leg connectivity
- Standard hardware holding the Hexapod together
- Modular compute architecture
- Deterministic real time compute provided by Xiao RP2350s
- ROS2 on Raspberry Pi 5 to allow easy build in to existing robotic development platforms

Price point: ~$2,000 USD.

Useful docs:

- Assembly Instructions: https://manufacturedmotion.github.io/Hex/Hex3/assembly-instructions/
- 3D printing guidelines: https://manufacturedmotion.github.io/Hex/Hex3/3d-printing/#print-recommendations
- Bill-of-Materials: https://docs.google.com/spreadsheets/d/1jHR1j3dNIgRCcadYxqtaftRhhn8tEC1XY8og_EQwXEU/edit?usp=sharing

Model files

base_individuals 7 files



base_core.3mf



base_pcb_guard.3mf



base_lower.3mf



base_lidar_mount.3mf



base_flat_x12.3mf



base_upper.3mf



base_electronics_mount.3mf



leg_individuals 22 files



s2_magnet_mount.3mf



s1_motor_bearing_mount.3mf



s1_encoder_cover.3mf



s3_bearing_side.3mf



s0_gear_cover_spacer.3mf



toe.3mf



s1_motor_bearing_holder.3mf



s2_connector_short.3mf



spring_aligner.3mf



s2_bearing_screw_mount.3mf



s1_bearing_holder_x2.3mf



s2_gear_side.3mf



s1_motor_cover.3mf



toe-tip.3mf



s3_gear_mount.3mf



s0_gear_cover.3mf



s1.3mf



s0.3mf



s2_bearing_side.3mf



s3_gear_side.3mf



s2_connector_long.3mf



s3_magnet_mount.3mf



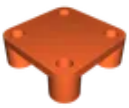
full_leg.3mf



base_parts.3mf

full_assembly_08_08_2026.f3z

encoder_insertion_tool.f3d



encoder_insertion_tool.step

full_assembly_08_08_2026.step

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